

Wanzheng Ning

Communication Engineering · Network Diagnostics · Mathematical Modeling · Mathematical Research

Education

Northeastern University at Qinhuangdao

Qinhuangdao, China

Undergraduate student in Communication Engineering

Research

Connected-Diagram Expansions for Hermite–Jensen Multipliers: All-Excess Summation and Critical Zero Asymptotics

Under review · Advances in Mathematics

Single-author work on connected-diagram expansions, all-excess summation, and critical zero asymptotics.

Critical Cubic Crossover from Hermite Universality in Riemann- ξ Jensen Polynomials

Under review · Journal of the London Mathematical Society

Single-author work on the critical cubic crossover from Hermite universality in Riemann- ξ Jensen polynomials.

Beyond Vertex Profiles: Edge-Local Spectral Information and Size-Dependent Scaling in Nonuniform Hypergraph Tensors

Submitted to Linear and Multilinear Algebra

First- and corresponding-author work with Qianzhi Ao as second author and four shared third authors, covering robust profile non-determination, edge-local tensor bounds, quotient reduction, and loose-star asymptotics.

Projects

NetSage

Kotlin · Jetpack Compose · DNS/TCP/TLS/HTTP

Built a serverless Android network-diagnostics workbench combining local log rules with active probes, explainable evidence, retesting, session history, and Markdown/JSON export.

CommLab

Python · Streamlit · wireless systems · edge intelligence

Built a local-first research platform with 130 interactive laboratory modes spanning physical-layer communications, ISAC, distributed learning, resilient edge inference, networked control, and robust offline policy evaluation.

Battery RUL and cascade utilization modeling

Python · survival modeling · graph optimization

Q1 to Q4 workflow on fully simulated data generated with semi-empirical assumptions, covering degradation stages, SOH/RUL, compatibility graphs, MILP grouping, and robust stress testing.

Australian high-speed rail design and management concept

Five-person carbody team · Proposed Design

Contributed to a lightweight composite carriage concept and independently developed a five-page front-end prototype for capital pooling, risk simulation, compliance workflow, and impact reporting using illustrative data.

Competitions

Mathematical modeling project

University-level Second Prize in the Mathematical Modeling Competition

Four-part battery reliability and utilization workflow on fully simulated data generated with semi-empirical assumptions; received the stated university-level second prize.

Scenic Guide Digital Human

Competition software project

Tourism guide prototype to which I contributed, with conversational and voice interaction, route guidance, narration, and knowledge management.

Technical skills

Network and mobile

Kotlin, Jetpack Compose, Android, network diagnostics, DNS/TLS/HTTP troubleshooting concepts

Modeling and research

Python, change-point regression, survival modeling, graph methods, mathematical optimization, asymptotic analysis, LaTeX

Software

FastAPI, JSON, Git

Languages

Chinese

GitHub accounts

lbrswne

ruoquecheng-eng